Weekly Radar
Anomaly Detection on Weekly Network Performance

Project 06 - Telco-Cloud Analytics Training

Information Technology Institute

Team Members

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Dataset: Performance.xlsx → weekly_region_carrier_tech.csv
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Dataset & Filtering

What was the dataset?

The source workbook was a **mobile network performance dataset** with **93,364 rows**. It mixed **daily and weekly records**, contained both **locality-level rows** and a **country aggregate**, and covered multiple **regions**, **carriers**, and **technologies**.

1) Removed daily rows

We kept **weekly rows only** so that all comparisons used the same time granularity.

2) Kept locality rows

The main workflow used **weekly locality rows** because they are the correct starting level for aggregation.

3) Separated country rows

Country aggregate rows were moved out of the main analysis table and kept only as benchmark rows.

4) Aggregated by region

Locality rows were aggregated into a clean **region × carrier × technology × week** table using **sample_count-weighted averages**.

5) Applied a sample threshold

We removed weak groups below **sample_count_total = 30** to keep reliability while preserving full regional coverage.

6) Added helper columns

Helper columns were added to support **mean vs. median** comparison and later anomaly analysis.

Preprocessing pipeline results

The table summarizes how the raw workbook was filtered and reduced into the final analysis dataset.

Processing stage	Rows
Raw spreadsheet rows	93,364
Weekly rows kept	58,858
Daily rows removed	34,506
Weekly locality rows	58,669
Weekly country rows	189
Aggregated region-carrier-technology-week rows	351
Rows removed by sample threshold (≥30)	8
Final analysis rows	**343**
Complete peer-comparison rows	330

Table 1: Row counts through the preprocessing pipeline.

EDA Highlights

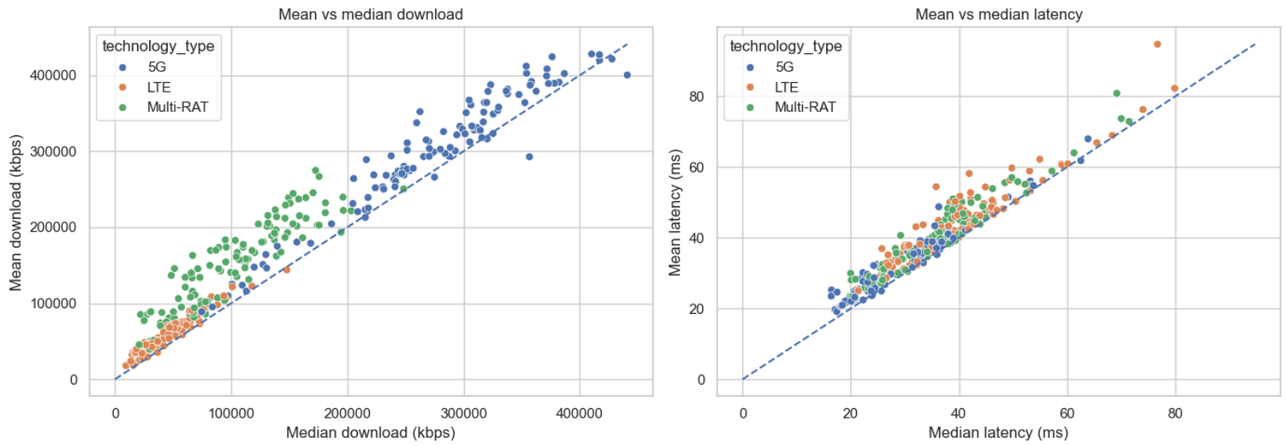

Figure 1: Mean-vs-median comparison for download and latency after cleaning.

Main EDA takeaway

- **Cleaning kept 343 analysis rows** after the sample threshold.
- Coverage stayed complete: **39/39 region-week** records remained for ranking and WoW analysis.
- The mean-vs-median plots show several skewed KPI distributions, especially in download speed.
- This supported using **median-based KPIs** and peer comparison instead of relying only on means.

EDA quality check

Threshold 30 gave the best balance: it removed the weakest aggregated rows but preserved all **13 regions × 3 weeks**. This avoided losing Q2/Q5 coverage while still improving reliability before anomaly detection.

Q1 Result - Which carrier is consistently below the regional average?

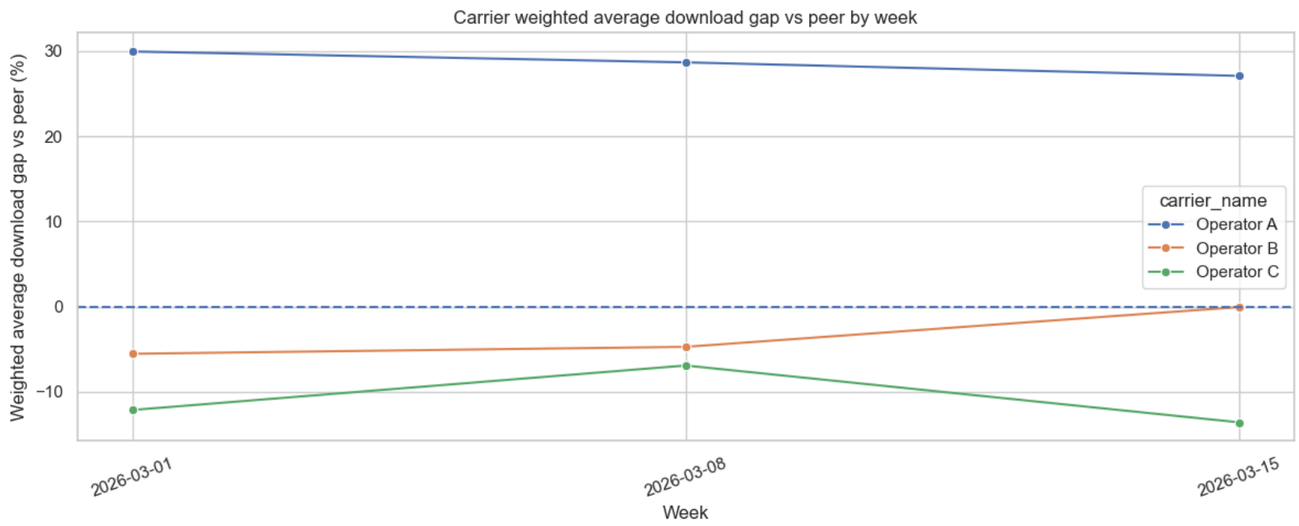

Figure 2: Average carrier download gap relative to regional peers across the three weeks.

Answer to Q1

Operator C was the only carrier that stayed **clearly below the regional peer average in all three weeks**. Operator A stayed clearly above peers, while Operator B was close to the baseline and not consistently majority-below.

Interpretation

This is not a one-week spike. Operator C had a repeated negative download gap and passed the majority-below-peer rule across all weeks.

How this graph was produced

- For each **week**, we computed each carrier's **weighted average download gap** across complete peer groups.
- Each carrier was compared with the **other carriers** in the same region, week, and technology.
- Decision rule: a carrier must be below peers in the **majority** of comparable groups across all weeks to be considered consistently weak.

Q2 Result - Which regions show the largest week-over-week changes?

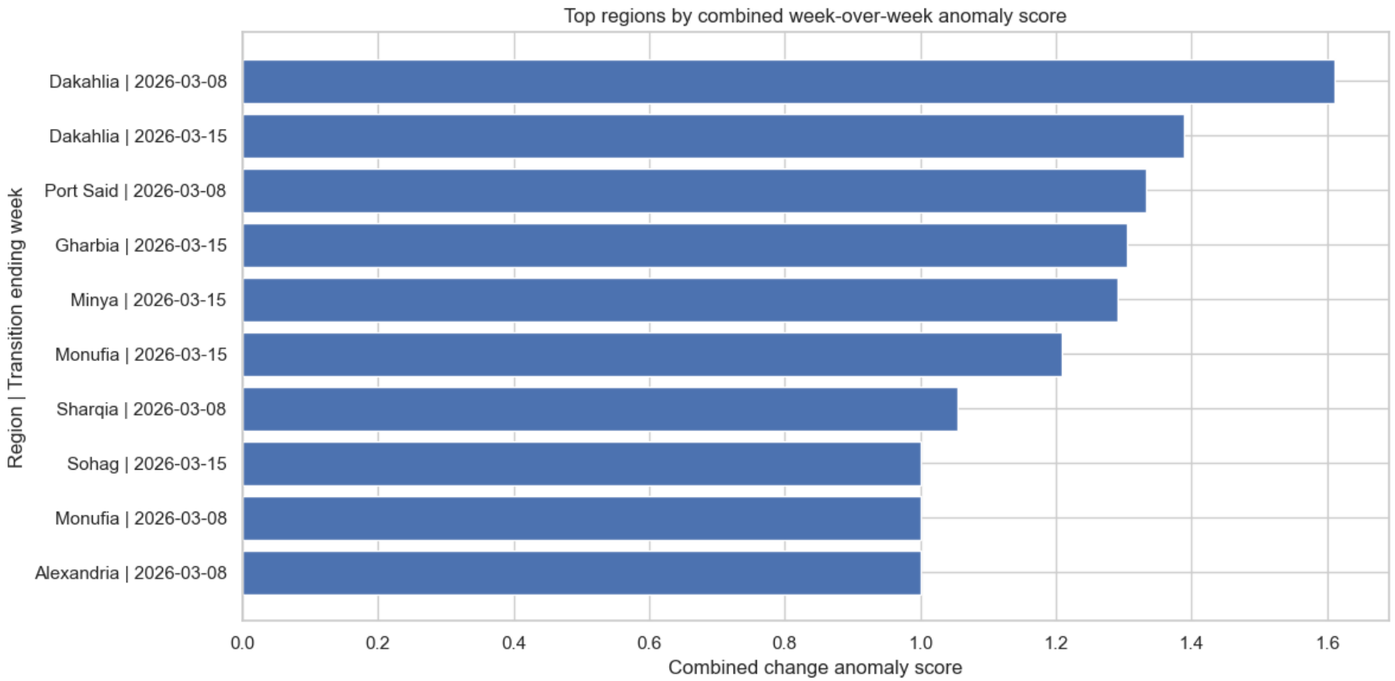

Figure 3: Top regions by combined week-over-week anomaly score across the 26 valid region transitions.

Answer to Q2

Largest **download drops**: **Assiut (-14.54%)**, **Monufia (-13.31%)**, and **Gharbia (-13.29%)**. Largest **latency rises**: **Monufia (+4.85%)**, **Sohag (+4.73%)**, and **Dakahlia (+4.08%)**. The combined WoW score flagged **3 region-week anomalies**: **Dakahlia** twice and **Port Said** once.

Week-over-week percentage change equation

$$\text{KPI WoW (\%)} = \frac{\text{Current Week Median KPI} - \text{Previous Week Median KPI}}{\text{Previous Week Median KPI}} \times 100$$

The 90th-percentile combined score was used because only 26 region transitions exist.

Q3 Result - Which region-carrier pairs are peer outliers?

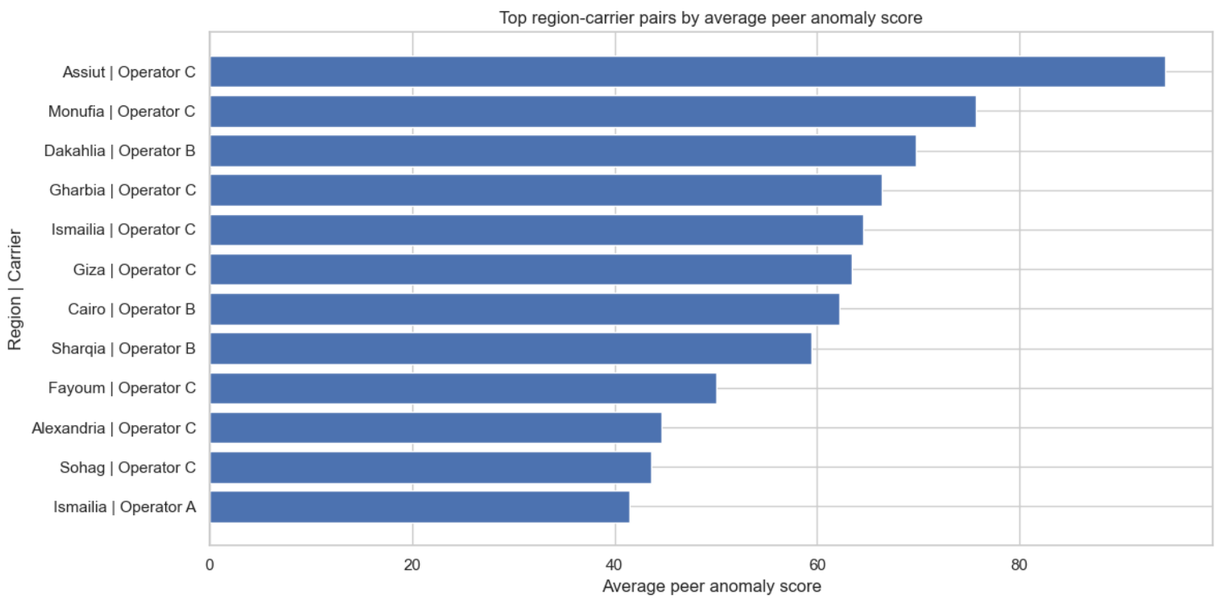

Figure 5: Top region-carrier pairs by average peer anomaly score.

Answer to Q3

Operator C dominated the highest average peer-anomaly scores, especially in **Assiut, Monufia, Gharbia, Ismailia,** and **Giza**. Operator B also appeared in **Dakahlia, Cairo,** and **Sharqia**, so the log was not blindly carrier-biased.

How this graph was produced

- Only **complete 3-carrier peer groups** were used: **330 rows** across **110 peer groups**.
- The peer-anomaly score penalizes **lower download** and **higher latency**; the 95th percentile produced **17 peer anomalies**.

$$\text{Cross-sectional Badness} = \max(0, -\text{Download Gap (\%)}) + \max(0, \text{Latency Gap (\%)})$$

Q4 & Q5 Results - Technology Comparison and Region Ranking

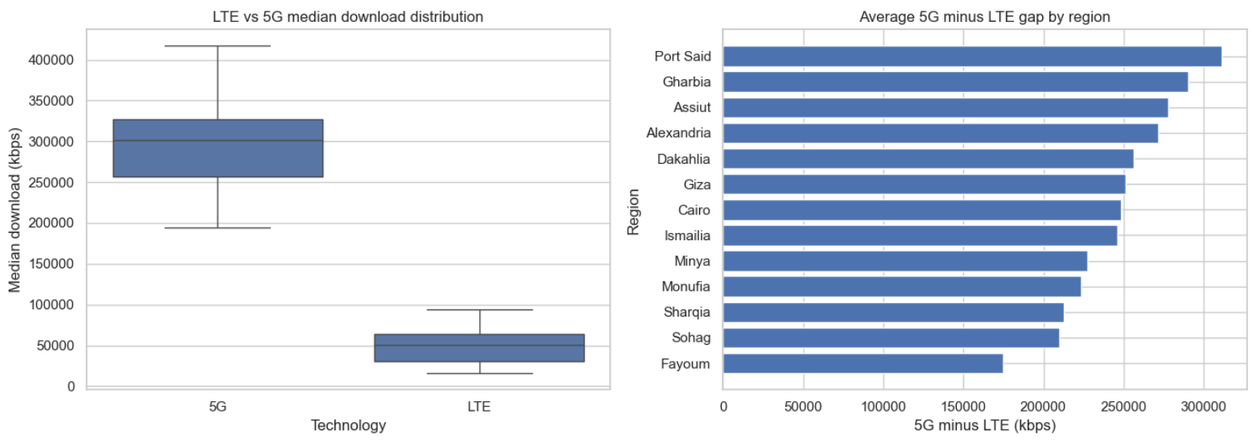

Figure 6: LTE vs 5G download distribution and 5G-minus-LTE regional gap.

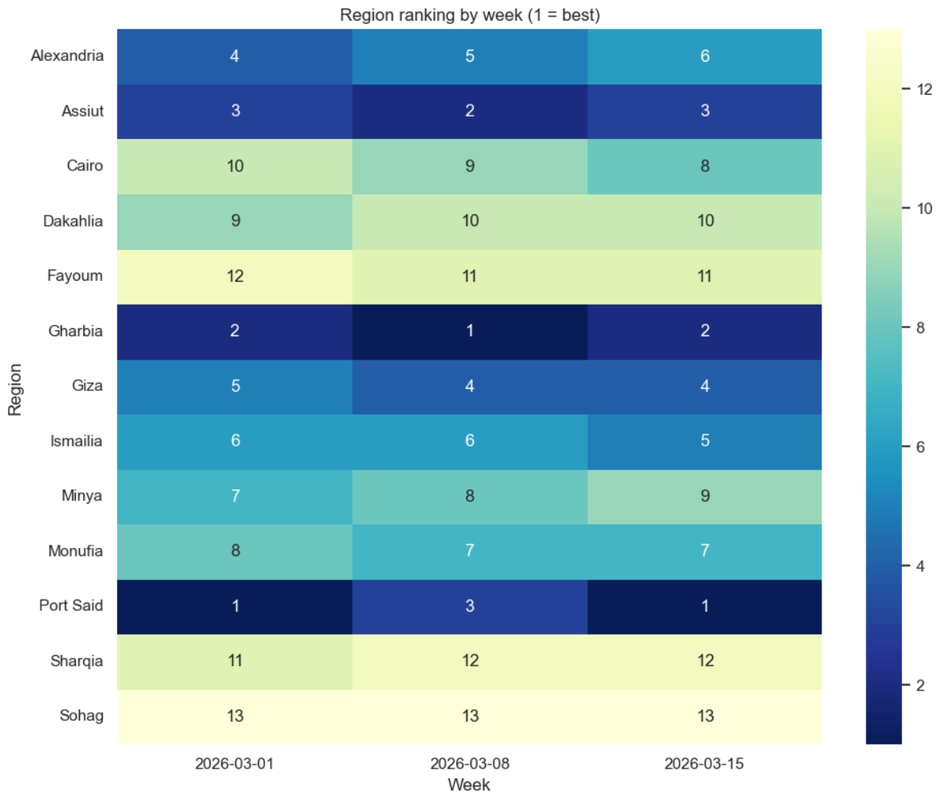

Figure 7: Region ranking by week (1 = best).

Answers

- **Q4:** Across the **39 valid region-week comparisons**, 5G download was higher than LTE in **39/39 cases**. The matched carrier check also passed: **109/109** carrier-region-week cases had 5G above LTE; average region-week median download was about **295 Mbps for 5G** versus **49 Mbps for LTE**.
- **Q5:** Ranking was built from **sample-count-weighted regional median download**; strongest regions were **Port Said, Gharbia,** and **Assiut**.
- **Stability: Port Said** had the largest movement (**1 → 3 → 1**), while **Sohag** stayed last (**13 → 13 → 13**).

Methods (P2 & P3) - How the anomaly detection was built

Method	What it does	Supports
Cross-sectional peer comparison	Compares each carrier against the regional peer average inside the same week and technology.	Q1, Q3
Week-over-week change analysis	Measures percentage change between consecutive weeks at region level to detect deterioration or improvement.	Q2
Technology distribution comparison	Compares LTE, Multi-RAT, and 5G download distributions across cleaned region-week records.	Q4
Region ranking analysis	Ranks all regions by region-week median download and tracks rank changes across the three weeks.	Q5

Table 2: Method design used to complete Project 06.

Why this method is appropriate

The dataset contains only **three weekly time points**, so long-history temporal models are not the best first choice. Final thresholds were: **sample_count_total ≥ 30**, complete **3-carrier peer groups**, **95th percentile** for peer anomalies, and **90th percentile** for WoW changes. Output logs contained **20 total anomalies**: **17 peer anomalies** and **3 week-over-week anomalies**.

Most Surprising Finding

Full Combined Anomaly Log

Type	Week	Region	Carrier	Tech	DL Gap/Δ	Lat Gap/Δ	Score
Peer	2026-03-08	Assiut	Operator C	LTE	-67.5%	+70.6%	1.983
Peer	2026-03-15	Assiut	Operator C	LTE	-68.5%	+66.8%	1.974
Peer	2026-03-01	Alexandria	Operator C	LTE	-65.2%	+57.9%	1.905
Peer	2026-03-08	Ismailia	Operator C	LTE	-61.7%	+69.1%	1.905
Peer	2026-03-15	Giza	Operator C	LTE	-65.8%	+42.3%	1.897
Peer	2026-03-01	Ismailia	Operator C	LTE	-61.9%	+64.4%	1.897
Peer	2026-03-01	Assiut	Operator C	LTE	-52.0%	+84.0%	1.871
Peer	2026-03-08	Giza	Operator C	LTE	-65.3%	+40.2%	1.871
Peer	2026-03-01	Monufia	Operator C	Multi-RAT	-52.7%	+37.7%	1.855
Peer	2026-03-15	Gharbia	Operator C	LTE	-56.6%	+63.2%	1.853
Peer	2026-03-01	Cairo	Operator B	Multi-RAT	-51.2%	+34.2%	1.838
Peer	2026-03-01	Giza	Operator C	LTE	-66.1%	+33.3%	1.836
Peer	2026-03-08	Monufia	Operator C	Multi-RAT	-65.4%	+22.5%	1.829
Peer	2026-03-15	Assiut	Operator C	5G	-24.7%	+67.9%	1.818
Peer	2026-03-15	Ismailia	Operator C	LTE	-51.1%	+57.5%	1.802
Peer	2026-03-15	Dakahlia	Operator B	Multi-RAT	-37.5%	+46.8%	1.786
Peer	2026-03-08	Dakahlia	Operator B	Multi-RAT	-38.0%	+44.3%	1.761
WoW	2026-03-08	Dakahlia	All carriers	All technologies	-11.0%	+2.9%	1.769
WoW	2026-03-15	Dakahlia	All carriers	All technologies	-6.3%	+4.1%	1.615
WoW	2026-03-08	Port Said	All carriers	All technologies	-7.3%	+2.7%	1.615

Table 3: Full anomaly log: 17 peer-comparison anomalies + 3 WoW region-change anomalies.

Most surprising result

Operator C still dominates the peer anomalies, but the full log also shows Operator B and regional WoW alerts.

- **20 total anomalies** were logged: **17 peer anomalies** and **3 week-over-week anomalies**.
- **Operator C appears in 14 of 17 peer anomalies**, showing a repeated carrier-technology weakness.
- **Operator B appears in 3 peer anomalies**, mainly Multi-RAT cases, so the method is not blindly carrier-biased.
- The **3 WoW alerts** are all-carrier regional changes: **Dakahlia twice** and **Port Said once**.

Lessons Learned

1) Preprocessing is part of the model

The biggest improvement came before anomaly detection: removing daily rows, separating country aggregates, aggregating localities with **sample-count weighting**, and choosing a balanced **threshold of 30**.

2) Cross-sectional comparison was the right fit

With only **3 weekly snapshots**, comparing each carrier against **same-week, same-region, same-technology peers** gives stronger evidence than forcing a time-series model. Complete peer groups gave **330 valid rows**.

3) Why not long-history models first?

- Rolling z-score / EWMA needs many historical weeks.
- LSTM or forecasting models need enough history to learn normal behavior.
- Example: with only 3 weeks, a sudden drop cannot be separated from normal seasonality.

4) Interpretability matters

A good anomaly record should answer: **which region, which carrier, which technology, which week, which KPI gap, and which threshold** caused the alert.

Key technical takeaway

Cross-sectional analysis is strongest when the dataset has **few time points but many peer entities**. Here, we had only 3 weeks but many comparable region-carrier-technology records, so peer comparison was more reliable than long-window temporal modeling. The final run kept **343 cleaned rows**, **330 peer-comparison rows**, and **39/39 region-week rows**. With 6--12+ months of weekly data, the next step would be adding rolling baselines, CUSUM/EWMA, or temporal anomaly models and validating them against known network incidents.